ELECTRONIC ANSWER BOOK

**I AGREE TO ABIDE BY THE SENATE POLICY ON CONDUCT OF ASSIGNMENTS, MID-TERM EXAM, AND FINAL EXAM**

**Instruction: Using Microsoft Word/WordPad/Notepad, type Student's name, Student ID number, type Chapter title and applicable Question Number for Assignment, Mid-Term Exam or Final Exam (as listed under Brightspace, Quizzes) before typing Solution below. One separate file for each Assignment/Mid-Term/Final Exam Question.**

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**Midterm – RBF – Question: 2a**

**Solution:**

Input:

| $k$ | $X_k$ | $d_k$ |
|---|---|---|
| 1 | [1, 0, 0] | 1 |
| 2 | [0, 1, 0] | 0 |
| 3 | [0, 0, 1] | -1 |

| $p$ | $T_p$ |
|---|---|
| 1 | [1, 0, 0] |
| 2 | [0, 0, 1] |

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Using the Gaussian functions, $y(X_k)$ becomes

$$y(X_k) = \sum_{p=1}^{2} w_p \exp\left(-\|X_k - T_p\|\right)^2 + w_0$$

To map the input to output patterns, we require,

$$y(X_k) = d_k \text{ for } k = 1, 2, 3$$

Where $X_k$ is an input vector and $d_k$ is the desired output

The radial basis function value at the $p$ hidden unit due to a $k$ th input pattern is equal to

$$g_{kp} = \exp\left(-\|X_k - T_p\|\right)^2 \text{ for } k = 1, 2, 3; p = 1, 2$$

Repeating the above procedures, the entries of the **G** matrix can be obtained as

$$G = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \\ g_{31} & g_{32} \end{bmatrix} = \begin{bmatrix} 1.0000 & 0.1353 \\ 0.1353 & 0.1353 \\ 0.1353 & 1.0000 \end{bmatrix}$$

Let the combined weight vector be,

$$W_c = [w_0, w_1, w_2]^T$$

The following matrix equation can be obtained

$$W_c = H^+ D$$

where $H^+$ represents the pseudo-inverse of the matrix $H$,

$$H = \begin{bmatrix} 1 & g_{11} & g_{12} \\ 1 & g_{21} & g_{22} \\ 1 & g_{31} & g_{32} \\ 1 & g_{41} & g_{42} \end{bmatrix} = \begin{bmatrix} 1.0000 & 1.0000 & 0.1353 \\ 1.0000 & 0.1353 & 0.1353 \\ 1.0000 & 0.1353 & 1.0000 \end{bmatrix}$$

Therefore,

$$H^+ = (H^T H)^{-1} H^T = \begin{bmatrix} -0.1565 & 1.3130 & -0.1565 \\ 1.1565 & -1.1565 & 0.0000 \\ -0.0000 & -1.1565 & 1.1565 \end{bmatrix}$$

$$W_c = H^+ D = \begin{bmatrix} -0.1565 & 1.3130 & -0.1565 \\ 1.1565 & -1.1565 & 0.0000 \\ -0.0000 & -1.1565 & 1.1565 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\boldsymbol{W_c} = \begin{bmatrix} \mathbf{0.0000} \\ \mathbf{1.1565} \\ \mathbf{-1.1565} \end{bmatrix}$$

Therefore, the sum of the 3 weight values is **0.0000**